THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 100°	Angle sum of a quadrilateral is 360°.
2	**(c)** 118°	Opposite angles are equal; adjacent angles add up to 180°.
3	**(a)** 44 cm	Opposite sides are equal.
4	**(c)** rhombus	Definition of a rhombus.
5	**(c)** 20 cm	$\sqrt{12^2+16^2}=20$.
6	**(b)** 4 cm	Midpoint theorem: $DE=\frac{1}{2}BC$.
7	**(a)** parallelogram	Definition.
8	**(b)** $12\sqrt{2}$ cm	$\sqrt{12^2+12^2}=12\sqrt{2}$.
9	**(d)** 55°	Angle sum of a quadrilateral is 360°.
10	**(b)** 72°	Opposite angles are equal; adjacent angles add up to 180°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle sum $=360^\circ$; Sum of the three given angles $=315^\circ$ (1); Fourth angle $=360^\circ-315^\circ=45^\circ$ (1).	1+1
12	$\angle C=\angle A=52^\circ$ (opposite angles); $\angle B=180^\circ-52^\circ=128^\circ$ (adjacent angles) (1); $\angle D=\angle B=128^\circ$ (1).	1+1
13	Opposite sides are equal: other sides 12 cm and 15 cm (1); Perimeter $=2(12+15)=54$ cm (1).	1+1
14	Angle sum $=360^\circ$; Sum of the three given angles $=260^\circ$ (1); Fourth angle $=360^\circ-260^\circ=100^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Diagonal $=\sqrt{9^2+12^2}$ (1); $=\sqrt{225}=15$ cm (1); Perimeter $=2(9+12)=42$ cm (1).	1+1+1
16	$EF=10$ cm, $FD=15$ cm, $DE=8$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $=10+15+8=33$ cm (2).	1+2
17	Angle sum $=360^\circ$ (1); Sum of the three given angles $=210^\circ$ (1); Fourth angle $=360^\circ-210^\circ=150^\circ$ (1).	1+1+1
	OR: Diagonal $=\sqrt{12^2+16^2}$ (1); $=\sqrt{400}=20$ cm (1); Perimeter $=2(12+16)=56$ cm (1).	1+1+1
18	$EF=15$ cm, $FD=14$ cm, $DE=5$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $=15+14+5=34$ cm (2).	1+2
	OR: Angle sum $=360^\circ$ (1); Sum of the three given angles $=270^\circ$ (1); Fourth angle $=360^\circ-270^\circ=90^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum of ratio terms $=12$ (1); One part $=360^\circ\div12=30^\circ$ (2); Angles: $30^\circ, 90^\circ, 150^\circ, 90^\circ$ (2).	1+2+2
	OR: Diagonals bisect each other at right angles (1); Side $=\sqrt{6^2+8^2}=10$ cm (1); Perimeter $=4\times10=40$ cm (1); Area $=\frac{1}{2}\times12\times16=96$ cm^2 (2).	1+1+1+2
20	Sum of ratio terms $=15$ (1); One part $=360^\circ\div15=24^\circ$ (2); Angles: $96^\circ, 24^\circ, 120^\circ, 120^\circ$ (2).	1+2+2
	OR: Diagonals bisect each other at right angles (1); Side $=\sqrt{12^2+16^2}=20$ cm (1); Perimeter $=4\times20=80$ cm (1); Area $=\frac{1}{2}\times24\times32=384$ cm^2 (2).	1+1+1+2

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE=2BC$ or $DE=\frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 48 cm	Opposite sides are equal.
2	**(b)** equal	Property.
3	**(c)** 16 cm	$\sqrt{20^2 - 12^2} = 16$.
4	**(a)** 15 cm	Midpoint theorem: $DE = \frac{1}{2}BC$.
5	**(b)** bisect each other at right angles	Property of a rhombus.
6	**(b)** $8\sqrt{2}$ cm	$\sqrt{8^2 + 8^2} = 8\sqrt{2}$.
7	**(d)** 70°	Angle sum of a quadrilateral is 360°.
8	**(c)** 128°	Opposite angles are equal; adjacent angles add up to 180°.
9	**(d)** 40 cm	Opposite sides are equal.
10	**(b)** rhombus	Property of rhombus.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle C = \angle A = 60^\circ$ (opposite angles); $\angle B = 180^\circ - 60^\circ = 120^\circ$ (adjacent angles) (1); $\angle D = \angle B = 120^\circ$ (1).	1+1
12	Opposite sides are equal: other sides 9 cm and 12 cm (1); Perimeter $= 2(9 + 12) = 42$ cm (1).	1+1
13	Angle sum $= 360^\circ$; Sum of the three given angles $= 305^\circ$ (1); Fourth angle $= 360^\circ - 305^\circ = 55^\circ$ (1).	1+1
14	$\angle C = \angle A = 70^\circ$ (opposite angles); $\angle B = 180^\circ - 70^\circ = 110^\circ$ (adjacent angles) (1); $\angle D = \angle B = 110^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Diagonal $= \sqrt{6^2 + 8^2}$ (1); $= \sqrt{100} = 10$ cm (1); Perimeter $= 2(6 + 8) = 28$ cm (1).	1+1+1
16	$EF = 7$ cm, $FD = 13$ cm, $DE = 13$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 7 + 13 + 13 = 33$ cm (2).	1+2
17	Angle sum $= 360^\circ$ (1); Sum of the three given angles $= 295^\circ$ (1); Fourth angle $= 360^\circ - 295^\circ = 65^\circ$ (1).	1+1+1
	OR: Diagonal $= \sqrt{18^2 + 24^2}$ (1); $= \sqrt{900} = 30$ cm (1); Perimeter $= 2(18 + 24) = 84$ cm (1).	1+1+1
18	$EF = 15$ cm, $FD = 13$ cm, $DE = 9$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 15 + 13 + 9 = 37$ cm (2).	1+2
	OR: Angle sum $= 360^\circ$ (1); Sum of the three given angles $= 250^\circ$ (1); Fourth angle $= 360^\circ - 250^\circ = 110^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum of ratio terms $= 15$ (1); One part $= 360^\circ \div 15 = 24^\circ$ (2); Angles: 96°, 120°, 24°, 120° (2).	1+2+2
	OR: Diagonals bisect each other at right angles (1); Side $= \sqrt{18^2 + 24^2} = 30$ cm (1); Perimeter $= 4 \times 30 = 120$ cm (1); Area $= \frac{1}{2} \times 36 \times 48 = 864$ cm^2 (2).	1+1+1+2
20	Sum of ratio terms $= 18$ (1); One part $= 360^\circ \div 18 = 20^\circ$ (2); Angles: 80°, 100°, 60°, 120° (2).	1+2+2
	OR: Diagonals bisect each other at right angles (1); Side $= \sqrt{9^2 + 12^2} = 15$ cm (1); Perimeter $= 4 \times 15 = 60$ cm (1); Area $= \frac{1}{2} \times 18 \times 24 = 216$ cm^2 (2).	1+1+1+2

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE = 2BC$ or $DE = \frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 8 cm	$\sqrt{10^2 - 6^2} = 8$.
2	**(c)** 20 cm	Midpoint theorem: $BC = 2DE$.
3	**(d)** rectangle	Each angle is 90°.
4	**(b)** $4\sqrt{2}$ cm	$\sqrt{4^2 + 4^2} = 4\sqrt{2}$.
5	**(a)** 50°	Angle sum of a quadrilateral is 360°.
6	**(d)** 64°	Opposite angles are equal; adjacent angles add up to 180°.
7	**(b)** 58 cm	Opposite sides are equal.
8	**(b)** rhombus and a rectangle	It has all properties of both.
9	**(b)** 10 cm	$\sqrt{6^2 + 8^2} = 10$.
10	**(b)** 5 cm	Midpoint theorem: $DE = \frac{1}{2}BC$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Opposite sides are equal: other sides 11 cm and 15 cm (1); Perimeter $= 2(11 + 15) = 52$ cm (1).	1+1
12	Angle sum $= 360^\circ$; Sum of the three given angles $= 280^\circ$ (1); Fourth angle $= 360^\circ - 280^\circ = 80^\circ$ (1).	1+1
13	$\angle C = \angle A = 64^\circ$ (opposite angles); $\angle B = 180^\circ - 64^\circ = 116^\circ$ (adjacent angles) (1); $\angle D = \angle B = 116^\circ$ (1).	1+1
14	Opposite sides are equal: other sides 7 cm and 10 cm (1); Perimeter $= 2(7 + 10) = 34$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Diagonal $= \sqrt{24^2 + 32^2}$ (1); $= \sqrt{1600} = 40$ cm (1); Perimeter $= 2(24 + 32) = 112$ cm (1).	1+1+1
16	$EF = 14$ cm, $FD = 6$ cm, $DE = 13$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 14 + 6 + 13 = 33$ cm (2).	1+2
17	Angle sum $= 360^\circ$ (1); Sum of the three given angles $= 310^\circ$ (1); Fourth angle $= 360^\circ - 310^\circ = 50^\circ$ (1).	1+1+1
	OR: Diagonal $= \sqrt{3^2 + 4^2}$ (1); $= \sqrt{25} = 5$ cm (1); Perimeter $= 2(3 + 4) = 14$ cm (1).	1+1+1
18	$EF = 10$ cm, $FD = 10$ cm, $DE = 6$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 10 + 10 + 6 = 26$ cm (2).	1+2
	OR: Angle sum $= 360^\circ$ (1); Sum of the three given angles $= 190^\circ$ (1); Fourth angle $= 360^\circ - 190^\circ = 170^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum of ratio terms $= 12$ (1); One part $= 360^\circ \div 12 = 30^\circ$ (2); Angles: 90°, 120°, 30°, 120° (2).	1+2+2
	OR: Diagonals bisect each other at right angles (1); Side $= \sqrt{24^2 + 32^2} = 40$ cm (1); Perimeter $= 4 \times 40 = 160$ cm (1); Area $= \frac{1}{2} \times 48 \times 64 = 1536$ cm^2 (2).	1+1+1+2
20	Sum of ratio terms $= 18$ (1); One part $= 360^\circ \div 18 = 20^\circ$ (2); Angles: 120°, 100°, 120°, 20° (2).	1+2+2
	OR: Diagonals bisect each other at right angles (1); Side $= \sqrt{3^2 + 4^2} = 5$ cm (1); Perimeter $= 4 \times 5 = 20$ cm (1); Area $= \frac{1}{2} \times 6 \times 8 = 24$ cm^2 (2).	1+1+1+2

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- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
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